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CLUTCH DEVICE

Abstract

A clutch device includes: a clutch outer rotatably journaled on a driving force transmission shaft and having an outer bottom portion formed in an annular shape and an outer cylindrical portion extending in the axial direction from an outer peripheral edge of the outer bottom portion; and a primary driven gear formed in an annular shape, assembled to the clutch outer, and receiving power from a power source. The outer bottom portion has a gear mounting portion inserted inside the primary driven gear and fitted to the primary driven gear in a relatively non-rotatable manner. The primary driven gear is formed from a first material. At least one of the outer bottom portion and the outer cylindrical portion is formed from a second material. The first material has higher strength than the second material.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] Priority is claimed on Japanese Patent Application No. 2024-029676, filed Feb. 29, 2024, the content of which is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a clutch device.

DESCRIPTION OF RELATED ART

[0003] In recent years, research and development on weight reduction that contributes to energy efficiency has been conducted in order to ensure that more people have access to affordable, reliable, sustainable, and advanced energy.

[0004] A clutch device may be provided in a torque transmission path between a power source such as an engine and a wheel. As a clutch device, there is a multiplate friction clutch. The multiplate friction clutch includes a cylindrical clutch outer that is rotatably journaled on an output shaft, a primary driven gear that is assembled to the clutch outer and receives power from a power source, a clutch center that is supported by the output shaft inside the clutch outer in a relatively non-rotatable manner, and friction members that are supported on the clutch outer and the clutch center in a relatively displaceable manner in the axial direction of the output shaft, and that exert a frictional force when contacting each other and transmit a rotational force from the clutch outer to the clutch center.

[0005] United States Patent Application, Publication No. 2020/0248755 discloses a constitution in which a primary driven gear is formed directly on the outer periphery of a clutch outer. In this constitution, the clutch device can be downsized in the axial direction as compared with a constitution in which the primary driven gear is disposed in axially overlapping relation with the clutch outer.

[0006] Meanwhile, in the present technology related to weight reduction, the clutch device has the following problems. If the primary driven gear is provided integrally with the clutch outer, the clutch outer also needs to be formed from a high-strength material such as iron in order to ensure the strength of the gear. This may increase the weight of the clutch device as compared with the case where the clutch outer is formed from aluminum.

[0007] In order to address the above problems, an object of the present application is to achieve weight reduction of a clutch device. This ultimately contributes to energy efficiency.

SUMMARY OF THE INVENTION

[0008] A clutch device according to a first aspect of the present invention includes: a clutch outer (**30**) that is rotatably journaled on an output shaft (**20**) and has an outer bottom portion (**31**) formed in an annular shape coaxial with the output shaft (**20**) and an outer cylindrical portion (**32**) extending in an axial direction of the output shaft (**20**) from an outer peripheral edge of the outer bottom portion (**31**); a primary driven gear (**50**) that is formed in an annular shape coaxial with the output shaft (**20**), is assembled to the clutch outer (**30**), and receives power from a power source; a clutch center (**60**) supported on the output shaft (**20**) in a relatively non-rotatable manner inside the clutch outer (**30**); and a friction member (**70**) supported on the clutch outer (**30**) and the clutch center (**60**) in a relatively displaceable manner in the axial direction of the output shaft (**20**), the friction member (**70**) having a plurality of friction plates (**71**,**72**), the friction member (**70**) exerting a frictional force through mutual contact between the plurality of friction plates (**71**,**72**) and transmitting a rotational force from the clutch outer (**30**) to the clutch center (**60**). The outer bottom portion (**31**) has a gear mounting portion (**37**) inserted inside the primary driven gear (**50**) and fitted to the primary driven gear (**50**) in a relatively non-rotatable manner. The primary driven gear (**50**) is formed from a first material. At least one of the outer bottom portion (**31**) and the outer

cylindrical portion (**32**) is formed from a second material. The first material has higher strength than the second material.
[0009] According to the first aspect, since the primary driven gear that receives power from the power source is formed from a material having higher strength than at least one of the outer bottom portion and the outer cylindrical portion, it is possible to select a material having a relatively low specific gravity as the material forming the clutch outer. Therefore, while ensuring the strength of the primary driven gear, the weight of the clutch outer can be reduced as compared with a case where the outer bottom portion and the outer cylindrical portion are formed from the same first material as the primary driven gear. Therefore, the weight of the clutch device can be reduced.
[0010] A clutch device according to a second aspect of the present invention is the clutch device according to the first aspect, in which the second material may have a lower specific gravity than the first material.
[0011] According to the second aspect compared with a case where the outer bottom portion and the outer cylindrical portion are formed from the same first material as the primary driven gear, the clutch outer can be made lighter, and thus the clutch device can be made lighter.
[0012] A clutch device according to a third aspect of the present invention is the clutch device according to the first aspect or the second aspect, in which the clutch outer (**30**) may have an outer shaft portion (**33**) connected to an inner peripheral edge of the outer bottom portion (**31**) and supported on the output shaft (**20**), and the outer shaft portion (**33**) may be formed from the first material.
[0013] According to the third aspect, the strength of the outer shaft portion can be ensured without increasing the size of the outer shaft portion, since the outer shaft portion of the clutch outer particularly requiring strength is formed from the first material.
[0014] A clutch device according to a fourth aspect of the present invention is the clutch device according to any one of the first to third aspects, in which the clutch outer (**30**) may have an outer shaft portion (**33**) connected to an inner peripheral edge of the outer bottom portion (**31**) and supported on the output shaft (**20**). The outer bottom portion (**31**) may have an outer peripheral connection portion (**35**) that engages with the primary driven gear (**50**) and is connected to the outer cylindrical portion (**32**) and a hub portion (**36**) that connects the outer peripheral connection portion (**35**) to the outer shaft portion (**33**). The hub portion (**36**) may be disposed between both ends of the primary driven gear (**50**) in the axial direction.
[0015] According to the fourth aspect, the hub portion can receive the load transmitted from the primary driven gear to the outer shaft portion when the primary driven gear receives power. As a result, the wall thickness of the hub portion and its surrounding area can be reduced as compared with a constitution in which the hub portion is axially displaced from the primary driven gear. Therefore, the increase in the weight of the clutch outer can be suppressed.
[0016] A clutch device according to a fifth aspect of the present invention is the clutch device according to the fourth aspect, further including a spigot-fit portion (**90**) where an inner peripheral portion (**53**) of the primary driven gear (**50**) and the gear mounting portion (**37**) are spigot-fitted to each other. The hub portion (**36**) may radially overlap the spigot-fit portion (**90**).
[0017] According to the fifth aspect, the hub portion can receive the load transmitted radially inward from the primary driven gear through the spigot-fit portion when the primary driven gear receives power. As a result, the wall thickness of the hub portion and its surrounding area can be reduced as compared with a constitution in which the hub portion is at a position not overlapping radially with the spigot-fit portion. Therefore, the increase in the weight of the clutch outer can be more effectively suppressed.
[0018] A clutch device according to a sixth aspect of the present invention is the clutch device according to any one of the first to fifth aspects, in which the clutch outer (**30**) may have an outer shaft portion (**33**) connected to an inner peripheral edge of the outer bottom portion (**31**) and inserted over the output shaft (**20**). The outer bottom portion (**31**) may have: an outer peripheral

connection portion (**35**) that engages with the primary driven gear (**50**) and is connected to the outer cylindrical portion (**32**); and a hub portion (**36**) that connects the outer peripheral connection portion (**35**) to the outer shaft portion (**33**). A connection portion between the hub portion (**36**) and the outer shaft portion (**33**) may be disposed between both ends of the primary driven gear (**50**) in the axial direction.
[0019] According to the sixth aspect, the load transmitted radially inward from the primary driven gear toward the outer shaft portion when the primary driven gear receives power can be efficiently released from the hub portion to the outer shaft portion. Therefore, it is possible to suppress the application of excessive load to the connection portion between the hub portion and the outer shaft portion.
[0020] A clutch device according to a seventh aspect of the present invention is the clutch device according to the sixth aspect, further including a spigot-fit portion (**90**) where an inner peripheral portion (**53**) of the primary driven gear (**50**) and the gear mounting portion (**37**) are spigot-fitted to each other. The connection portion between the hub portion (**36**) and the outer shaft portion (**33**) may radially overlap the spigot-fit portion (**90**).
[0021] According to the seventh aspect, the load transmitted radially inward from the primary driven gear through the spigot-fit portion when the primary driven gear receives power can be efficiently released from the hub portion to the outer shaft portion by the connection portion between the hub portion and the outer shaft portion. Therefore, it is possible to more effectively suppress the application of excessive load to the connection portion between the hub portion and the outer shaft portion.
[0022] A clutch device according to an eighth aspect of the present invention is the clutch device according to any one of the first to seventh aspects, in which the gear mounting portion (**37**) may have an outer peripheral portion (**370**) with a tooth portion (**38**) that meshes with an inner peripheral portion (**53**) of the primary driven gear (**50**), and the outer bottom portion (**31**) may be formed with a lightening hole (**43**) along the tooth portion (**38**).
[0023] According to the eighth aspect, the hub portion is more likely to deflect around the lightening hole. As a result, the load transmitted from the primary driven gear to the hub portion when the primary driven gear receives power can be dispersed by the deflection of the hub portion. Therefore, it is possible to suppress the concentration of load in a specific location and reduce the wall thickness of the hub portion and its surrounding area.
[0024] A clutch device according to a ninth aspect of the present invention is the clutch device according to any one of the first to eighth aspects, further including a fastening member (**85**) that fixes the primary driven gear (**50**) and the clutch outer (**30**) to each other. The gear mounting portion (**37**) may have an outer peripheral portion (**370**) with a tooth portion (**38**) that meshes with an inner peripheral portion (**53**) of the primary driven gear (**50**). The fastening member (**85**) may be disposed radially inward of an outer peripheral end of the tooth portion (**38**).
[0025] According to the ninth aspect, the fastening member can be provided so as not to axially overlap the teeth on the outer periphery of the primary driven gear. Therefore, it is possible to suppress an increase in the outer diameter of the primary driven gear.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] FIG. **1** is a cross-sectional view illustrating a portion of a power unit according to an embodiment;
[0027] FIG. **2** is a cross-sectional view of a clutch outer and a primary driven gear according to the embodiment;
[0028] FIG. **3** is a side view of a clutch device according to the embodiment as viewed from the

inside in the axial direction, with a pressing plate removed;
[0029] FIG. **4** is a perspective view of the clutch device according to the embodiment as viewed from the inside in the axial direction; and
[0030] FIG. **5** is an enlarged perspective view illustrating the periphery of a gear mounting portion in the clutch outer according to the embodiment.
DETAILED DESCRIPTION OF THE INVENTION
[0031] Hereinafter, an embodiment of the present invention will be described with reference to the drawings. Note that, in the following description, constitutions having the same or similar functions are denoted by the same reference numbers. In addition, redundant descriptions of these constitutions may be omitted. Also, in the following description, directions such as front, rear, up, down, left, and right coincide with directions with respect to a vehicle described below. That is, the left-right direction coincides with the vehicle width direction. Also, in the drawings used in the following description, arrow LH indicates the left side.
[0032] FIG. **1** is a cross-sectional view illustrating a portion of a power unit according to an embodiment.
[0033] As illustrated in FIG. **1**, a clutch device **7** according to the present embodiment is provided integrally with an engine **3** and a transmission **5** as a portion of a power unit **1** mounted on a saddle-ride vehicle such as a motorcycle. The engine **3** includes a crankshaft **10** extending in the vehicle width direction and a crankcase **11** that houses the crankshaft **10**. The crankshaft **10** rotates by converting the reciprocating motion of the piston into rotational motion. The crankcase **11** is made of metal and is formed by fastening a left crankcase **11**L and a right crankcase **11**R, which are divided into right and left halves, to each other.
[0034] The transmission **5** is housed in a rear portion of the crankcase **11**. The rear portion of the crankcase **11** also serves as a transmission case **11***a* that houses the transmission **5**. The transmission **5** is a stepped transmission having a main shaft **12** and a countershaft **13** rotatably supported in the transmission case **11***a*, and a transmission gear group **14** extending over the main shaft **12** and the countershaft **13**. The main shaft **12** and the countershaft **13** extend in the vehicle width direction. The transmission gear group **14** switches between the gear pairs used for power transmission between the main shaft **12** and the countershaft **13** in the transmission gear group **14** by movement of the shift fork of a change mechanism (not illustrated). The clutch device **7** is coupled to the right end of the main shaft **12**. The rotational power of the crankshaft **10** is transmitted to the main shaft **12** via the clutch device **7** and is transmitted from the main shaft **12** to the countershaft **13** via a gear pair of the transmission gear group **14**. The countershaft **13** constitutes the output shaft of the power unit **1**. The countershaft **13** protrudes to the left of the transmission case **11***a* and is coupled to a drive sprocket **15**. The rotation of the countershaft **13** is transmitted from the left side of the transmission case **11***a* to a rear wheel via a chain drive power transmission mechanism.
[0035] A clutch cover **16** is coupled to the transmission case **11***a*. The clutch cover **16** is disposed on the right side of the transmission case **11***a* and is coupled to the right crankcase **11**R. The clutch cover **16** is disposed on an extension of the main shaft **12**. The clutch cover **16** defines a clutch chamber with the right crankcase **11**R.
[0036] The clutch device **7** is a multiplate friction clutch that connects and disconnects the power transmission between the crankshaft **10** of the engine **3** and the main shaft **12** of the transmission **5**. The clutch device **7** is engaged and disengaged by the operation of a clutch operating element by a driver or the driving of an actuator.
[0037] The clutch device **7** is disposed between the transmission case **11***a* and the clutch cover **16**. The clutch device **7** includes a driving force transmission shaft **20** that is provided as the output shaft of the clutch device **7** and rotatably supported by the crankcase **11**, a clutch outer **30** that is journaled on the driving force transmission shaft **20** in a relatively rotatable manner, a primary driven gear **50** that is assembled to the clutch outer **30** and receives power from the engine **3**, which

is a power source, and a clutch center **60** that is supported in a relatively non-rotatable manner by the driving force transmission shaft **20** inside the clutch outer **30**.

[0038] The clutch outer **30**, the primary driven gear **50**, and the clutch center **60** are provided so as to be rotatable about the rotation axis of the driving force transmission shaft **20**. In the following description, the rotation axis of the driving force transmission shaft **20** is referred to as a common axis O. In addition, the direction along the common axis O is referred to as the axial direction, the direction orthogonal to the common axis O and extending radially from the common axis O is referred to as the radial direction, and the direction circling around the common axis O is referred to as the circumferential direction. In the present embodiment, the axial direction coincides with the vehicle width direction, the axial inside is the direction toward the center of the vehicle along the vehicle width direction, and the axial outside is the direction away from the center of the vehicle along the vehicle width direction.

[0039] The driving force transmission shaft **20** is the main shaft **12** of the transmission **5**. The driving force transmission shaft **20** is rotatably supported by the right crankcase **11**R via a ball bearing **17**A and is rotatably supported by the left crankcase **11**L via a ball bearing **17**B. The driving force transmission shaft **20** extends axially outward from the right crankcase **11**R.

[0040] The primary driven gear **50** rotates synchronously with the crankshaft **10** by meshing with a primary drive gear **19** that rotates integrally with the crankshaft **10** of the engine **3**. The primary driven gear **50** is formed in an annular shape (see FIG. **4**). The primary driven gear **50** is disposed so as to overlap the clutch outer **30** from the inside in the axial direction. The entire primary driven gear **50** is disposed inward of the contour of the clutch outer **30** as viewed in the axial direction (see FIG. **3**). The primary driven gear **50** is formed from a first material. The first material will be described later.

[0041] FIG. **2** is a cross-sectional view of the clutch outer and primary driven gear according to the embodiment. FIG. **3** is a side view of the clutch device according to the embodiment as viewed from the inside in the axial direction, with a pressing plate and a disc spring removed.

[0042] As illustrated in FIGS. **2** and **3**, the primary driven gear **50** has a gear outer peripheral portion **51** on which external teeth that mesh with the teeth of the primary drive gear **19** are formed, and a gear inner peripheral portion **53** that is fitted to the clutch outer **30** in a relatively non-rotatable manner. The gear inner peripheral portion **53** includes an inner flange **54** extending radially inward from the inner peripheral surface of the gear outer peripheral portion **51** and extending around the entire circumference, and engaging teeth **55** protruding radially inward from the inner peripheral edge of the inner flange **54**. The inner flange **54** is connected to the gear outer peripheral portion **51** in axially spaced relation to both ends in the axial direction on the inner peripheral surface of the gear outer peripheral portion **51**. The engaging teeth **55** are formed with protrusions **55***a* protruding from their tooth tips along the axial direction toward the clutch outer **30** side. The plurality of engaging teeth **55** are provided around the entire circumference. The plurality of engaging teeth **55** are arranged at equally spaced intervals in the circumferential direction, except where rivets **85** described later are arranged.

[0043] FIG. **4** is a perspective view of the clutch device according to the embodiment as viewed from the inside in the axial direction.

[0044] As illustrated in FIGS. **2** and **4**, the clutch outer **30** is formed in a bottomed cylindrical shape as a whole. The clutch outer **30** includes an annular outer bottom portion **31**, an outer cylindrical portion **32** extending axially outward from the outer peripheral edge of the outer bottom portion **31**, and an outer shaft portion **33** connected to the inner peripheral edge of the outer bottom portion **31**. The outer bottom portion **31** and the outer cylindrical portion **32** are integrally formed. The outer bottom portion **31** and the outer cylindrical portion **32** are formed from a second material. The second material has a lower strength than the first material. Note that in the present embodiment, the strength compared between the first material and the second material is tensile strength. The second material has a lower specific gravity than the first material. In the present embodiment, the first material is aluminum, and the second material is iron. The outer cylindrical portion **32** is formed with fitting grooves **32***a* open in the inner peripheral surface thereof. The plurality of fitting grooves **32***a* are formed at equally spaced intervals in the circumferential direction.

[0045] As illustrated in FIGS. **2** and **3**, the outer bottom portion **31** overlaps the primary driven gear **50** from the axial outside. The outer bottom portion **31** includes an outer peripheral connection portion **35** that engages with the primary driven gear **50** and is connected to the outer cylindrical portion **32**, and an outer hub portion **36** that connects the outer peripheral connection portion **35** to the outer shaft portion **33**. The outer peripheral connection portion **35** is formed in an annular shape. The outer peripheral connection portion **35** includes a gear mounting portion **37**. The gear mounting portion **37** is provided on the inner periphery of the outer peripheral connection portion **35** and is formed in an annular shape. The gear mounting portion **37** protrudes further inward in the axial direction than the outer periphery of the outer peripheral connection portion **35** and extends around the entire circumference of the outer shaft portion **33**. The gear mounting portion **37** is inserted inside the primary driven gear **50**. The gear mounting portion **37** is inserted inside the gear inner peripheral portion **53** of the primary driven gear **50** and is fitted to the primary driven gear **50** in a relatively non-rotatable manner.

[0046] The gear mounting portion **37** has an outer peripheral portion **370** and an inner peripheral portion **37***i*. The outer peripheral portion **370** overlaps the inner flange **54** of the primary driven gear **50** from the axial outside and is fitted inside the gear outer peripheral portion **51** of the primary driven gear **50**. Thus, the outer peripheral portion **370** of the gear mounting portion **37** and the primary driven gear **50** are spigot-fitted to each other. Hereinafter, the location where the gear mounting portion **37** and the primary driven gear **50** are spigot-fitted to each other will be referred to as a spigot-fit portion **90**. The inner peripheral portion **37***i* protrudes further inward in the axial direction than the outer peripheral portion **370**. The inner peripheral portion **37***i* is disposed inside the gear inner peripheral portion **53** of the primary driven gear **50**.

[0047] FIG. **5** is an enlarged perspective view illustrating the periphery of the gear mounting portion in the clutch outer according to the embodiment.

[0048] As illustrated in FIGS. **3** and **5**, the inner peripheral portion **37***i* includes engaged teeth **38** that mesh with the engaging teeth **55** of the primary driven gear **50**. The engaged teeth **38** and the engaging teeth **55** are fitted to each other by radial recesses and protrusions. Thus, the inner peripheral portion **37***i* of the gear mounting portion **37** and the primary driven gear **50** are engaged with each other in a relatively non-rotatable manner. The plurality of engaged teeth **38** are provided around the entire circumference so as to correspond to the engaging teeth **55**. The tooth grooves of the engaged teeth **38** are open on the axially inner end surface of the inner peripheral portion **37***i*. Recesses **39** into which the protrusions **55***a* of the engaging teeth **55** are inserted are formed in the axially inner end surface of the gear mounting portion **37**. The recesses **39** include the tooth grooves of the engaged teeth **38** when viewed from the axial direction. In the illustrated example, the recesses **39** are provided so as to extend across the inner peripheral portion **37***i* and outer peripheral portion **370** of the gear mounting portion **37**. However, the recesses may be provided only in the inner peripheral portion **37***i* of the gear mounting portion **37**.

[0049] As illustrated in FIGS. **2** and **3**, a fastening seat **41** to which the rivet **85** (fastening member) is attached is provided on the outer bottom portion **31** in such a manner as to be raised inward in the axial direction. The fastening seat **41** is provided in the gear mounting portion **37**. A plurality of the fastening seats **41** are provided at equally spaced intervals in the circumferential direction. In the illustrated example, five fastening seats **41** are provided. The fastening seats **41** are arranged inside the primary driven gear **50** as viewed from the axial direction. The fastening seats **41** are located at positions corresponding to the inner peripheral portion **37***i* of the gear mounting portion **37** in the radial direction. The fastening seats **41** are each formed with a through hole which penetrates the outer bottom portion **31** in the axial direction and into which the rivet **85** is inserted from the inside in the axial direction. The rivet **85** includes a head **85***a* located axially inside the fastening seat **41**. The head **85***a* of the rivet **85** is disposed radially inward of the outer peripheral ends of the engaged teeth **38**. The rivet **85** has a pressing plate **86** sandwiched between the head **85***a* and the fastening seat **41**. The pressing plate **86** is formed in an annular plate shape coaxial with the primary driven gear **50**. The pressing plate **86** overlaps the engaging teeth **55** of the primary driven gear **50** when viewed from the axial direction. The pressing plate **86** presses the primary driven gear **50** toward the outer bottom portion **31** side (axially outward) via disc springs **87** to restrict the primary driven gear **50** from falling off. Note that since the power from the engine **3** does not act directly on the rivets **85**, it is sufficient if the number and load of the rivets **85** are selected so that the rivets **85** can evenly press the disk springs **87**.

[0050] As illustrated in FIGS. **2** and **4**, the outer hub portion **36** is formed in an annular shape. The outer hub portion **36** is formed in a tabular shape extending orthogonal to the axial direction. The outer peripheral edge of the outer hub portion **36** is connected to the inner peripheral portion **37***i* of the gear mounting portion **37**. The outer hub portion **36** is located between both ends of the gear mounting portion **37** in the axial direction. The outer hub portion **36** radially overlaps the spigot-fit portion **90** between the gear mounting portion **37** and the primary driven gear **50**. Further, the outer hub portion **36** overlaps in the radial direction the meshing part between the engaging teeth **55** of the primary driven gear **50** and the engaged teeth **38** of the gear mounting portion **37**. A lightening hole **43** is formed in the outer hub portion **36**. The lightening hole **43** penetrates the outer hub portion **36** in the axial direction. A plurality of the lightening holes **43** are formed at equally spaced intervals in the circumferential direction. In the present embodiment, five lightening holes **43** are formed. The lightening holes **43** are formed so as to avoid the fastening seats **41** in the circumferential direction. The lightening holes **43** are formed in the outer periphery of the outer hub portion **36** along the engaged teeth **38**.

[0051] The outer shaft portion **33** is formed in a cylindrical shape. The outer shaft portion **33** is inserted over the driving force transmission shaft **20** and rotatably supported by the driving force transmission shaft **20**. The outer shaft portion **33** is coupled to the inner peripheral edge of the outer hub portion **36**. Thus, the clutch outer **30** is provided as a single member. The outer shaft portion **33** is formed from a first material. The outer shaft portion **33** is joined to the outer hub portion **36** by welding. For example, the outer shaft portion **33** is coupled to the outer hub portion **36** by ring mash welding. The connection portion between the outer shaft portion **33** and the outer hub portion **36** is disposed between both ends of the primary driven gear **50** in the axial direction. The connection portion between the outer shaft portion **33** and the outer hub portion **36** radially overlaps the spigot-fit portion **90** between the gear mounting portion **37** and the primary driven gear **50**. Further, the outer hub portion **36** overlaps in the radial direction the meshing part between the engaging teeth **55** of the primary driven gear **50** and the engaged teeth **38** of the gear mounting portion **37**.

[0052] In the present embodiment, there is no shock-absorbing mechanism between the clutch outer **30** and the primary driven gear **50** to absorb torque fluctuations transmitted from one of the clutch outer **30** and the primary driven gear **50** to the other, and the clutch outer **30** and the primary driven gear **50** always rotate together. Furthermore, in the present embodiment, the clutch outer **30** and the primary driven gear **50** can rotate together by the mesh of the engaging teeth **55** and the engaged teeth **38**, and unlike the conventional coupling structure between the primary driven gear and the clutch outer, the primary driven gear **50** is not provided with a through hole into which a boss formed on the clutch outer is inserted.

[0053] As illustrated in FIG. **1**, the clutch center **60** is disposed inside the outer cylindrical portion **32**. The clutch center **60** is formed in an annular shape when viewed from the axial direction. The clutch center **60** includes a center hub portion **61** provided on the inner periphery of the clutch center **60** and spline-fitted to the outer peripheral surface of the driving force transmission shaft **20**, an annular pressure-receiving wall portion **62** provided on the outer periphery of the clutch center

60, and a center cylindrical portion **63** extending axially outward from the inner peripheral edge of the pressure-receiving wall portion **62**. Note that in the present embodiment, the center hub portion **61**, the pressure-receiving wall portion **62**, and the center cylindrical portion **63** are formed as separate members, but it is sufficient if the center hub portion **61**, the pressure-receiving wall portion **62**, and the center cylindrical portion **63** can rotate together. The center hub portion **61** is fastened to the driving force transmission shaft **20** by a nut screwed to the right end of the driving force transmission shaft **20**.

[0054] The pressure-receiving wall portion **62** is adjacent to and axially faces the outer bottom portion **31** of the clutch outer **30**. The pressure-receiving wall portion **62** has a pressure-receiving surface **62***a* facing axially outward. The pressure-receiving surface **62***a* is an annular flat surface extending along a direction orthogonal to the axial direction.

[0055] The center cylindrical portion **63** is disposed inside the outer cylindrical portion **32**. The center cylindrical portion **63** is disposed to face the outer cylindrical portion **32** in the radial direction. The center cylindrical portion **63** is formed with an engaged protrusion **64**. The engaged protrusion **64** protrudes outward in the radial direction. The engaged protrusion **64** extends in the axial direction. A plurality of the engaged protrusions **64** are provided at equally spaced intervals in the circumferential direction.

[0056] The clutch device **7** further includes a friction plate set **70** (friction member) that is supported on the clutch outer **30** and the clutch center **60** in a relatively displaceable manner in the axial direction and that exerts a frictional force through mutual contact between plates and transmits a rotational force from the clutch outer **30** to the clutch center **60**, and a pressure member **75** that is provided so as to be displaceable with respect to the clutch center **60** between a first position where the mutual contact between the plates of the friction plate set **70** is restrained and a second position where the mutual contact between the plates of the friction plate set **70** is released.

[0057] The friction plate set **70** is disposed axially outward with respect to the pressure-receiving wall portion **62**. The friction plate set **70** is formed entirely in a cylindrical shape and is disposed coaxially with the common axis O. The friction plate set **70** is disposed between the outer cylindrical portion **32** of the clutch outer **30** and the center cylindrical portion **63** of the clutch center **60**. The friction plate set **70** is in contact with the pressure-receiving surface **62***a* of the pressure-receiving wall portion **62**. The friction plate set **70** is made up of friction plates **71** and clutch plates **72** stacked so that the friction plates **71** and clutch plates **72** overlap alternately in the axial direction. The friction plates **71** and the clutch plates **72** are each formed in an annular shape and arranged coaxially with the common axis O.

[0058] Engagement outer protrusions **71***a* protruding radially outward are provided on the outer periphery of the friction plates **71**. The engagement outer protrusions **71***a* protrude further outward in the radial direction than the outer peripheral edges of the clutch plates **72**. The engagement outer protrusions **71***a* fit into the fitting grooves **32***a* of the outer cylindrical portion **32**. Thus, the friction plates **71** are fitted to the inner peripheral surface of the outer cylindrical portion **32** so as to be axially slidable and rotate together with the clutch outer **30**.

[0059] Engagement inner protrusions **72***a* protruding radially inward are provided on the inner periphery of the clutch plates **72**. The engagement inner protrusions **72***a* protrude further inward in the radial direction than the inner peripheral edges of the friction plates **71**. The engagement inner protrusions **72***a* fit into grooves between the engaged protrusions **64** provided on the center cylindrical portion **63**. Thus, the clutch plates **72** are spline-fitted to the outer peripheral surface of the center cylindrical portion **63** so as to be axially slidable and rotate together with the clutch center **60**.

[0060] When the mutual contact between the friction plates **71** and the clutch plates **72** is established and the friction plate set **70** is frictionally engaged as one unit, the rotational force is transmitted from the clutch outer **30** to the clutch center **60** through the friction plate set **70**. When the friction plates **71** and the clutch plates **72** are released from mutual contact, slippage occurs between the friction plates **71** and the clutch plates **72**, and the transmission of the rotational force is disconnected.

[0061] The pressure member **75** is disposed axially outward with respect to the clutch center **60**. The pressure member **75** axially sandwiches the friction plate set **70** with the pressure-receiving wall portion **62** from the opposite side (axial outside) from the clutch center **60** in the axial direction. When in the first position in the axial direction, the pressure member **75** sandwiches the friction plate set **70** with the clutch center **60** and generates a frictional force between the friction plates **71** and the clutch plates **72** to frictionally engage the friction plate set **70** as one unit. The pressure member **75** is displaced axially outward from the first position toward the second position to reduce the frictional force generated between the friction plates **71** and the clutch plates **72**, thereby releasing the frictional engagement of the friction plate set **70**.

[0062] The pressure member **75** is formed in an annular shape when viewed from the axial direction, and is disposed coaxially with the common axis O. The pressure member **75** includes a pressure wall portion **76** provided on the outer periphery thereof. The pressure wall portion **76** faces the friction plate set **70** from the opposite side from the pressure-receiving wall portion **62** of the pressure member **75** in the axial direction. The pressure wall portion **76** has a pressure surface **76***a* facing axially inward. The pressure surface **76***a* is an annular flat surface extending along a direction orthogonal to the axial direction. The pressure surface **76***a* contacts the friction plate set **70** from the opposite side from the pressure-receiving surface **62***a* of the clutch center **60**. The pressure member **75** is biased toward the first position by a spring member (not illustrated). The pressure member **75** is supported at its center by a lifter member **81** described later. The pressure member **75** is pressed axially outward via the lifter member **81** by a push rod (not illustrated) disposed in the driving force transmission shaft **20**.

[0063] As described above, the clutch device **7** according to the present embodiment includes the clutch outer **30** having the outer bottom portion **31** and the outer cylindrical portion **32** extending in the axial direction from the outer peripheral edge of the outer bottom portion **31**, and the primary driven gear **50** assembled to the clutch outer **30** and receiving power from the power source. The outer bottom portion **31** has the gear mounting portion **37** inserted inside the primary driven gear **50** and fitted to the primary driven gear **50** in a relatively non-rotatable manner. The primary driven gear **50** is formed from a first material. The outer bottom portion **31** and the outer cylindrical portion **32** are formed from a second material. The first material has higher strength than the second material. With this constitution, since the primary driven gear **50** that receives power from the power source is formed from a material having higher strength than the outer bottom portion **31** and the outer cylindrical portion **32**, it is possible to select a material having a relatively low specific gravity as the material forming the clutch outer **30**. Therefore, while ensuring the strength of the primary driven gear **50**, the weight of the clutch outer **30** can be reduced as compared with a case where the outer bottom portion and the outer cylindrical portion are formed from the same first material as the primary driven gear **50**. In the present embodiment, as the strength of the primary driven gear **50**, the tensile strength of the material of the primary driven gear **50** is made higher than the tensile strength of the material of the outer bottom portion **31** and the outer cylindrical portion **32**, so that durability against the stress generated at the tooth root of the primary driven gear **50** can be ensured. Therefore, the weight of the clutch device **7** can be reduced.

[0064] The second material has a lower specific gravity than the first material. With this constitution compared with a case where the outer bottom portion and the outer cylindrical portion are formed from the same first material as the primary driven gear **50**, the clutch outer **30** can be made lighter, and thus the clutch device **7** can be made lighter.

[0065] The clutch outer **30** has the outer shaft portion **33** connected to the inner peripheral edge of the outer bottom portion **31** and supported by the output shaft. The outer shaft portion **33** is formed from a first material. With this constitution, since the outer shaft portion **33** of the clutch outer **30** particularly requiring strength is formed from the first material, the strength of the outer shaft portion **33** can be ensured without increasing the size of the outer shaft portion **33**.

[0066] The clutch outer **30** has the outer shaft portion **33** connected to the inner peripheral edge of the outer bottom portion **31** and supported by the output shaft. The outer bottom portion **31** has the outer peripheral connection portion **35** that engages with the primary driven gear **50** and is connected to the outer cylindrical portion **32** and the outer hub portion **36** that connects the outer peripheral connection portion **35** to the outer shaft portion **33**. The outer hub portion **36** is disposed between both ends of the primary driven gear **50** in the axial direction. With this constitution, the outer hub portion **36** can receive the load transmitted from the primary driven gear **50** to the outer shaft portion **33** when the primary driven gear **50** receives power. As a result, the wall thickness of the outer hub portion **36** and its surrounding area can be reduced as compared with a constitution in which the outer hub portion is axially displaced from the primary driven gear **50**. Therefore, the increase in the weight of the clutch outer **30** can be suppressed.

[0067] The outer hub portion **36** radially overlaps the inner peripheral portion **37***i* of the primary driven gear **50** and the spigot-fit portion **90** of the gear mounting portion **37**. With this constitution, the outer hub portion **36** can receive the load transmitted radially inward from the primary driven gear **50** through the spigot-fit portion **90** when the primary driven gear **50** receives power. As a result, the wall thickness of the outer hub portion **36** and its surrounding area can be reduced as compared with a constitution in which the outer hub portion is at a position not overlapping the spigot-fit portion **90** in the radial direction. Therefore, the increase in the weight of the clutch outer **30** can be more effectively suppressed.

[0068] The connection portion between the outer hub portion **36** and the outer shaft portion **33** is disposed between both ends of the primary driven gear **50** in the axial direction. With this constitution, the load transmitted radially inward from the primary driven gear **50** toward the outer shaft portion **33** when the primary driven gear **50** receives power can be efficiently released from the outer hub portion **36** to the outer shaft portion **33**. Therefore, it is possible to suppress the application of excessive load to the connection portion between the outer hub portion **36** and the outer shaft portion **33**.

[0069] The connection portion between the outer hub portion **36** and the outer shaft portion **33** radially overlaps the inner peripheral portion **37***i* of the primary driven gear **50** and the spigot-fit portion **90** of the gear mounting portion **37**. With this constitution, the load transmitted radially inward from the primary driven gear **50** through the spigot-fit portion **90** when the primary driven gear **50** receives power can be efficiently released from the outer hub portion **36** to the outer shaft portion **33** by the connection portion between the outer hub portion **36** and the outer shaft portion **33**. Therefore, it is possible to more effectively suppress the application of an excessive load to the connection portion between the outer hub portion **36** and the outer shaft portion **33**.

[0070] The outer peripheral portion **370** of the gear mounting portion **37** has the engaged teeth **38** that mesh with the engaging teeth **55** of the inner peripheral portion **37***i* of the primary driven gear **50**. The outer bottom portion **31** is formed with the lightening hole **43** along the engaged teeth **38**. With this constitution, the outer hub portion **36** is more likely to deflect around the lightening hole **43**. As a result, the load transmitted from the primary driven gear **50** to the outer hub portion **36** when the primary driven gear **50** receives power can be dispersed by the deflection of the outer hub portion **36**. Therefore, it is possible to suppress the concentration of load in a specific location and reduce the wall thickness of the outer hub portion **36** and its surrounding area.

[0071] The clutch device **7** further includes the rivet **85** that fixes the primary driven gear **50** and the clutch outer **30** to each other. The rivet **85** is disposed radially inward of the outer peripheral ends of the engaged teeth **38**. With this constitution, the rivet **85** can be provided so as not to overlap axially with the teeth on the outer periphery of the primary driven gear **50**. Therefore, it is possible to suppress an increase in the outer diameter of the primary driven gear **50**. However, the clutch device may include a screw as the fastening member instead of the rivet **85**. If the primary driven gear and the clutch outer are fixed to each other by the screw, the replacement of the

primary driven gear becomes easier, and therefore the primary driven gear can be replaced with another one as necessary, such as to change the number of teeth.
[0072] Note that the present invention is not limited to the embodiment that has been described above with reference to the drawings, and various modifications are conceivable within the technical scope.
[0073] For example, in the above embodiment, an example in which the present invention is applied to a power unit with an engine as a power source has been described, but the present invention may also be applied to a power unit with a motor as a power source.
[0074] In the above embodiment, both the outer bottom portion **31** and the outer cylindrical portion **32** of the clutch outer **30** are formed from the second material, but the present invention is not limited to this constitution. That is, it is sufficient if at least one of the outer bottom portion and the outer cylindrical portion may be formed from the second material. Furthermore, in the above embodiment, the outer shaft portion **33** is formed from the same first material as the primary driven gear **50**, but the material forming the outer shaft portion is not limited, and may be formed from, for example, the second material.
[0075] In the above embodiment, the engaging teeth **55** of the primary driven gear **50** are arranged in the circumferential direction, but the present invention is not limited to this constitution. It is sufficient if the primary driven gear and the gear mounting portion of the clutch outer are engaged at least at one location in the circumferential direction in a relatively non-rotatable manner by radial recesses and protrusions.
[0076] Additionally, it is possible to replace the constituent elements in the above-described embodiment with well-known constituent elements as appropriate without departing from the gist of the present invention.
BRIEF DESCRIPTION OF THE REFERENCE SYMBOLS
[0077] **7** clutch device [0078] **20** driving force transmission shaft (output shaft) [0079] **30** clutch outer [0080] **31** outer bottom portion [0081] **32** outer cylindrical portion [0082] **33** outer shaft portion [0083] **35** outer peripheral connection portion [0084] **36** outer hub portion (hub portion) [0085] **37** gear mounting portion [0086] **38** engaged teeth (tooth portion) [0087] **43** lightening hole [0088] **50** primary driven gear [0089] **60** clutch center [0090] **70** friction plate set (friction member) [0091] **71** friction plate (friction plate) [0092] **72** clutch plate (friction plate) [0093] **85** rivet (fastening member) [0094] **90** spigot-fit portion

Claims

1. A clutch device comprising: a clutch outer that is rotatably journaled on an output shaft and has an outer bottom portion formed in an annular shape coaxial with the output shaft and an outer cylindrical portion extending in an axial direction of the output shaft from an outer peripheral edge of the outer bottom portion; a primary driven gear that is formed in an annular shape coaxial with the output shaft, is assembled to the clutch outer, and receives power from a power source; a clutch center supported on the output shaft in a relatively non-rotatable manner inside the clutch outer; and a friction member supported on the clutch outer and the clutch center in a relatively displaceable manner in the axial direction of the output shaft, the friction member having a plurality of friction plates, the friction member exerting a frictional force through mutual contact between the plurality of friction plates and transmitting a rotational force from the clutch outer to the clutch center, wherein the outer bottom portion has a gear mounting portion inserted inside the primary driven gear and fitted to the primary driven gear in a relatively non-rotatable manner, the primary driven gear is formed from a first material, at least one of the outer bottom portion and the outer cylindrical portion is formed from a second material, and the first material has higher strength than the second material.
2. The clutch device according to claim 1, wherein the second material has a lower specific gravity

than the first material.

3. The clutch device according to claim 1, wherein the clutch outer has an outer shaft portion connected to an inner peripheral edge of the outer bottom portion and supported on the output shaft, and the outer shaft portion is formed from the first material.

4. The clutch device according to claim 1, wherein the clutch outer has an outer shaft portion connected to an inner peripheral edge of the outer bottom portion and supported on the output shaft, and the outer bottom portion has an outer peripheral connection portion that engages with the primary driven gear and is connected to the outer cylindrical portion and a hub portion that connects the outer peripheral connection portion to the outer shaft portion, the hub portion being disposed between both ends of the primary driven gear in the axial direction.

5. The clutch device according to claim 4, further comprising a spigot-fit portion where an inner peripheral portion of the primary driven gear and the gear mounting portion are spigot-fitted to each other, wherein the hub portion radially overlaps the spigot-fit portion.

6. The clutch device according to claim 1, wherein the clutch outer has an outer shaft portion connected to an inner peripheral edge of the outer bottom portion and inserted over the output shaft, the outer bottom portion has an outer peripheral connection portion that engages with the primary driven gear and is connected to the outer cylindrical portion and a hub portion that connects the outer peripheral connection portion to the outer shaft portion, and a connection portion between the hub portion and the outer shaft portion is disposed between both ends of the primary driven gear in the axial direction.

7. The clutch device according to claim 6, further comprising a spigot-fit portion where an inner peripheral portion of the primary driven gear and the gear mounting portion are spigot-fitted to each other, wherein the connection portion between the hub portion and the outer shaft portion radially overlaps the spigot-fit portion.

8. The clutch device according to claim 1, wherein the gear mounting portion has an outer peripheral portion with a tooth portion that meshes with an inner peripheral portion of the primary driven gear, and the outer bottom portion is formed with a lightening hole along the tooth portion.

9. The clutch device according to claim 1, further comprising a fastening member that fixes the primary driven gear and the clutch outer to each other, wherein the gear mounting portion has an outer peripheral portion with a tooth portion that meshes with an inner peripheral portion of the primary driven gear, and the fastening member is disposed radially inward of an outer peripheral end of the tooth portion.